

Day 7 — TNPSC Group 2A Study Material

Part A – General Studies: Life Science: Classification, Evolution, Genetics, Physiology, Nutrition, Health & Hygiene, Human Diseases

READING MATERIAL

This is the broadest clause in Unit I, spanning classification, genetics, physiology, nutrition, and disease. Real exam evidence favors mismatch-style and match-the-following questions over plain definitions, so several items below are framed that way.

Q: Real exam: Which of these genetic-disorder pairings is a MISMATCH? Down's Syndrome–Autosomal trisomy of 21st chromosome; Haemophilia–Sex Linked; Klinefelter Syndrome–XO Complement; Turner Syndrome–Female with retarded growth.

A: Klinefelter Syndrome–XO Complement is the mismatch. Klinefelter Syndrome is actually XXY (an extra X chromosome in males); XO (a missing X chromosome) is Turner Syndrome, which correctly presents as female with retarded growth — that pairing in the list is actually correct, the error is specifically in mislabeling Klinefelter as XO.

Q: Real exam: What are minisatellites, in DNA terms?

A: Short repeating DNA sequences consisting of 10-60 base pairs (bp) long base pairs — used in DNA profiling/fingerprinting.

Q: Real exam: Which statements about antibiotics are true — (i) active against a large number of microorganisms, (ii) enhance fungal growth, (iii) minimum inhibitory concentration is effective against different microorganisms?

A: (i) and (iii) are true. (ii) is false — antibiotics target bacteria, they don't 'enhance' fungal growth (and many antibiotics are themselves derived FROM fungi, e.g. Penicillin).

Q: What is Streptomycin active against, and where does Penicillin come from?

A: Streptomycin is active against Gram-negative bacteria. Penicillin is an antibiotic obtained from a blue-green mold (Penicillium, which is itself classified as a Fungus).

Q: What causes liver cirrhosis, and what category of drug is nicotine?

A: Cirrhosis of the liver is caused by chronic (long-term) alcohol intake. Nicotine belongs to the Stimulants group of drugs.

Q: What bacterium is most associated with stomach ulcers, and what disease does Bacillus anthracis cause?

A: Helicobacter pylori causes most ulcers. Bacillus anthracis causes Anthrax in cattle and humans.

Q: How does Typhoid spread, and what is Dengue's transmission cycle?

A: Typhoid spreads through contamination of drinking water and milk. Dengue follows a Man→Mosquito→Man transmission cycle.

Q: Which plasma protein is used in antibody formation?

A: Globulin (not albumin, fibrinogen, or thrombin — those have other blood-plasma roles).

Q: What first-aid combination should be given to an electrocuted person with cardiac arrest?

A: All of the above: external cardiac compression massage, mouth-to-mouth resuscitation, AND cardio-pulmonary resuscitation (CPR) together — not just one of these alone.

Q: Name a fresh-water habitat example and the driest habitat type, per basic classification.

A: Fresh water habitat: a pond. Driest habitat: deserts.

Q: What is the main function of stomata in plants, and what are the primary organs of photosynthesis?

A: Stomata's main function is transpiration (water vapour release/gas exchange). The primary organs of photosynthesis are leaves.

MOCK TEST (WITH ANSWERS)

1. Which of the following genetic-disorder pairings is a MISMATCH?

A. Down's Syndrome – Autosomal trisomy of 21st chromosome

B. Haemophilia – Sex Linked

C. Klinefelter Syndrome – XO Complement ✓

D. Turner Syndrome – Female with retarded growth

Explanation: Klinefelter Syndrome is XXY, not XO — XO is Turner Syndrome. [hard, question-bank-2025]

2. Minisatellites are DNA sequences consisting of base pairs in which approximate range?

A. 1-5 bp

B. 10-60 bp ✓

C. 100-200 bp

D. 500-1000 bp

Explanation: 10-60 base pairs long repeating sequences. [medium, question-bank-2026]

3. Which statement about antibiotics is FALSE?

A. They are active against a large number of microorganisms

B. Minimum inhibitory concentration is effective against different microorganisms

C. They enhance the growth of fungi ✓

D. All of the above are true

Explanation: Antibiotics do not enhance fungal growth — many are even derived from fungi. [medium, question-bank-2026]

4. Streptomycin is mainly active against which type of bacteria?

A. Gram-positive bacteria

B. Gram-negative bacteria ✓

C. Viruses

D. Fungi

Explanation: Gram-negative bacteria. [medium, web-research]

5. Chronic intake of which substance most commonly causes cirrhosis of the liver?

A. Tobacco

B. Alcohol ✓

C. Cocaine

D. Opium

Explanation: Alcohol. [easy, web-research]

6. Penicillin, a widely used antibiotic, is obtained from:

A. A bacterium

B. A blue-green mold (fungus) ✓

C. A virus

D. A protozoan

Explanation: Penicillium, a fungus (blue-green mold). [easy, web-research]

7. Ulcers are mostly caused by infection from which bacterium?

A. Streptococci

B. Staphylococci

C. Helicobacter pylori ✓

D. Mycobacterium tuberculosis

Explanation: Helicobacter pylori. [medium, web-research]

8. Typhoid is most commonly spread through:

A. Mosquito bites

B. Contaminated drinking water and milk ✓

C. Airborne droplets

D. Direct skin contact

Explanation: Contaminated water/milk. [easy, web-research]

9. Dengue virus follows which transmission cycle?

A. Man-Man

B. Man-Birds-Man

C. Man-Mosquito-Man ✓

D. Man-Pig-Man

Explanation: Man-Mosquito-Man. [easy, web-research]

10. Which plasma protein plays the key role in antibody formation?

A. Albumin

B. Globulin ✓

C. Fibrinogen

D. Thrombin

Explanation: Globulin. [medium, web-research]

11. Which of the following is the main function of stomata in a plant leaf?

A. Photosynthesis

B. Transpiration ✓

C. Absorption of minerals

D. Storage of starch

Explanation: Transpiration (water vapour/gas exchange). [easy, web-research]

12. A pond is a classic example of which type of habitat?

A. Marine habitat

B. Fresh water habitat ✓

C. Desert habitat

D. Mountain habitat

Explanation: Fresh water habitat. [easy, web-research]

Part B – Aptitude & Mental Ability: Compound Interest

READING MATERIAL

Compound Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC tests the standard CI formula, a useful SI-vs-CI shortcut, AND disguises CI as 'successive percentage change' word problems (depreciation, year-over-year growth) — recognize all three shapes.

Q: Formula: What is the Compound Interest formula, and the shortcut for the DIFFERENCE between SI and CI over 2 years?

A: Amount = $P(1+R/100)^T$, so CI = Amount – P. For exactly 2 years, there's a shortcut: $CI - SI = P \times (R/100)^2$. This avoids computing CI and SI separately when you only need their difference.

Q: Real exam (2022): Find the difference between SI and CI for 2 years on a principal of Rs.5,000 at 8% per annum.

A: Using the shortcut: difference = $P \times (R/100)^2 = 5000 \times (0.08)^2 = 5000 \times 0.0064 = \text{Rs.}32$. (Full check: SI = $5000 \times 8 \times 2/100 = 800$. CI = $5000 \times (1.08^2 - 1) = 5000 \times 0.1664 = 832$. Difference = $832 - 800 = 32$ ✓.)

Q: Real exam (2022): Find the compound interest on Rs.5,000 at 12% per annum for 2 years, compounded annually.

A: CI = $P \times ((1+R/100)^T - 1) = 5000 \times (1.12^2 - 1) = 5000 \times (1.2544 - 1) = 5000 \times 0.2544 = \text{Rs.}1,272$.

Q: Real exam (2025): A sum becomes Rs.13,380 after 3 years and Rs.20,070 after 6 years on compound interest. Find the original sum.

A: The amount after 6 years equals the amount after 3 years multiplied by the SAME growth factor again (since compounding is multiplicative): $(1+R/100)^3 = 20070/13380 = 1.5$. The sum itself: $P = \text{Amount-after-3-years} \div (1+R/100)^3 = 13380 \div 1.5 = \text{Rs.}8,920$.

Q: Real exam (2024): A motorcycle's value 2 years ago was Rs.80,000. It depreciates 4% in the first year, then increases 4% in the second year. Find the present value.

A: Apply each year's change as its own multiplier, in sequence — this is compound interest with a changing/mixed rate: $80,000 \times 0.96$ (first year, -4%) $\times 1.04$ (second year, +4%) = $80,000 \times 0.9984 = \text{Rs.}79,872$. Note the result is slightly LESS than the original Rs.80,000 — a decrease followed by an equal-percentage increase never fully cancels out, since the increase applies to a smaller base.

MOCK TEST (WITH ANSWERS)

1. Find the difference between SI and CI for 2 years on a principal of Rs.8,000 at 5% per annum.

- A. Rs.18
- B. Rs.20 ✓**
- C. Rs.22
- D. Rs.25

Explanation: Difference = $P \times (R/100)^2 = 8000 \times (0.05)^2 = 8000 \times 0.0025 = \text{Rs.}20$. [medium, authored]

2. Find the compound interest on Rs.10,000 at 10% per annum for 2 years, compounded annually.

- A. Rs.2,000
- B. Rs.2,100 ✓**
- C. Rs.2,200
- D. Rs.1,900

Explanation: CI = $10000 \times (1.10^2 - 1) = 10000 \times 0.21 = \text{Rs.}2,100$. [easy, authored]

3. A sum becomes Rs.16,000 after 2 years and Rs.19,360 after 4 years on compound interest. Find the original sum.

- A. Rs.12,500
- B. Rs.13,000
- C. Rs.13,223 ✓**
- D. Rs.14,000

Explanation: $(1+R/100)^2 = 19360/16000 = 1.21$. $P = 16000/1.21 \approx \text{Rs.}13,223$. [hard, authored]

4. A machine valued at Rs.50,000 depreciates by 10% in the first year and a further 10% in the second year. Find its value after 2 years.

- A. Rs.40,000
- B. Rs.40,500 ✓**
- C. Rs.41,000
- D. Rs.39,500

Explanation: $50,000 \times 0.90 \times 0.90 = 50,000 \times 0.81 = \text{Rs.}40,500$. [medium, authored]

5. Find the compound interest on Rs.20,000 at 15% per annum for 2 years, compounded annually.

A. Rs.6,000

B. Rs.6,450 ✓

C. Rs.6,200

D. Rs.6,500

Explanation: $CI = 20000 \times (1.15^2 - 1) = 20000 \times 0.3225 = \text{Rs.}6,450$. [medium, authored]

Part C – General English: Poem: "Wandering Singers" — Sarojini Naidu

READING MATERIAL

A Sarojini Naidu poem about folk musicians who travel from place to place, detached from worldly ties, singing of history, beauty, and fate. Real coaching-site evidence tests theme, in-context meaning, and figure-of-speech together.

Q: What is the central theme of "Wandering Singers"?

A: A wandering life and detachment — the singers move freely from place to place (compared to birds, butterflies, and clouds), unattached to any single home, spreading songs of love, beauty, history, and fate.

Q: What do the singers' songs cover, thematically?

A: Glorious cities whose past has been forgotten; the past lives of noble, happy, and beautiful women; and mighty battles fought by brave kings — in short, beauty, history, glory, bravery, happiness, and sorrow.

Q: What does "The voice of the wind is the voice of our fate" mean, and what figure of speech is it?

A: It means they follow their destiny wherever it leads — like the wind, their path isn't fixed or self-chosen. It's a Metaphor (fate is directly equated with the wind's voice, not compared using 'like/as').

Q: What does "All men are our kindred, the world is our home" imply about the singers?

A: They feel at home anywhere — they treat all people as family and don't see themselves as tied to one fixed home.

Q: What instrument do the wandering singers carry?

A: A lute.

Q: Who wrote "Wandering Singers"?

A: Sarojini Naidu.

MOCK TEST (WITH ANSWERS)

1. "Wandering Singers" was written by:

- A. Rabindranath Tagore
- B. Sarojini Naidu ✓**
- C. Sarala Devi
- D. Subramania Bharati

Explanation: Sarojini Naidu. [easy, web-research]

2. The central theme of the poem is:

- A. Modern music
- B. Freedom of travel
- C. Sacrifice and patriotism
- D. Wandering life and detachment ✓**

Explanation: Wandering life and detachment. [medium, web-research]

3. "The voice of the wind is the voice of our fate" means:

- A. Wind can be controlled
- B. They control their destiny
- C. They follow destiny wherever it leads ✓**
- D. They fear the wind

Explanation: They follow destiny wherever it leads, just as the wind's direction isn't chosen. [medium, web-research]

4. "The voice of the wind is the voice of our fate" uses which figure of speech?

- A. Simile
- B. Personification
- C. Metaphor ✓**
- D. Alliteration

Explanation: Metaphor — fate is directly equated with the wind's voice. [medium, web-research]

5. "All men are our kindred, the world is our home" implies that the singers:

A. Live in houses

B. Feel at home anywhere ✓

C. Dislike strangers

D. Stay only in forests

Explanation: They feel at home anywhere, treating all people as family. [medium, web-research]

6. What instrument do the wandering singers carry?

A. Harp

B. Lute ✓

C. Drum

D. Flute

Explanation: A lute. [easy, web-research]